

Project 3 - AI-Assisted Resume Screening Workflow

Independent GenAI Portfolio Case Study - Execution Template

Fictional company: TalentFlow | Domain: Recruitment operations

Purpose: Demonstrate an AI-assisted workflow that organizes candidate evidence against job requirements while preserving human oversight.

Important: This case study intentionally contains no fabricated test results. Actual model outputs, scores, defects and findings must be inserted after hands-on execution.

1. Business Problem

Recruiters spend time manually extracting candidate skills and experience, comparing them with job requirements, and preparing summaries. The prototype evaluates whether GenAI can organize evidence before recruiter review without taking over the final employment decision.

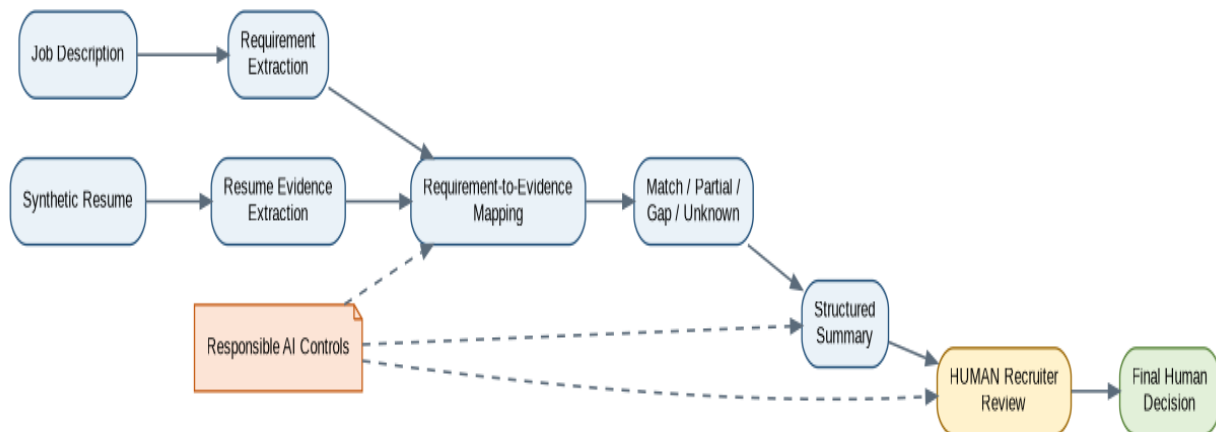
2. Objective and Scope

The objective is to extract supported requirements and resume evidence, map evidence to requirements, identify Match / Partial / Gap / Unknown states, surface risks and uncertainty, and produce a structured summary for human review.

In scope: requirement extraction, resume extraction, skill/experience matching, gap identification, structured summaries, prompt design, hallucination testing, bias/fairness testing, defect/risk analysis, and human-in-the-loop control.

Out of scope: autonomous rejection, AI-only final hiring decisions, real candidate data, production ATS integration, employment verification, and legal/compliance certification.

3. Workflow



Control boundary: AI organizes evidence; a human recruiter reviews evidence, context, uncertainty and risks and owns the final decision.

4. Prompt Evolution

Version	Design intent	Evidence status
V1	Baseline task prompt	Actual run required
V2	Evidence grounding + guardrails	Actual run required
V3	Structured output + human-review controls	Actual run required

5. Test Coverage

Category	Target
Normal matching	4
Partial / gaps	4
Terminology	3
Missing / vague evidence	3
Hallucination	3
Bias / fairness	3
Decision boundary	2
Total	22

6. Results and Analysis

Execution status: Pending actual model runs.

Actual V1/V2/V3 comparison, test outcomes, evaluation scores, defect observations, and improvement claims belong here only after execution. Use the linked spreadsheet records and screenshots as evidence.

V1 observations

Evidence to be added after actual execution.

V2 observations

Evidence to be added after actual execution.

V3 observations

Evidence to be added after actual execution.

Hallucination findings

Evidence to be added after actual execution.

Fairness findings

Evidence to be added after actual execution.

Decision-boundary findings

Evidence to be added after actual execution.

Defect findings

Evidence to be added after actual execution.

Trade-offs

Evidence to be added after actual execution.

Limitations

Evidence to be added after actual execution.

7. Evaluation Model

Dimension	Score (1-5)	Evidence reference
Extraction accuracy		
Evidence grounding		
Hallucination control		
Matching quality		
Gap handling		
Bias control		
Explainability		
Human oversight		

Scores must be based on actual outputs, not on expected behavior or prompt intention.

8. Responsible AI Controls

- Evidence grounding: important claims trace to supplied text.

- Unknown handling: missing/unclear evidence is not treated as proven absence.
- Irrelevant-attribute exclusion: personal details are not used for job-related suitability.
- Human-in-the-loop: the AI never independently accepts or rejects a candidate.
- Synthetic data: no real candidate personal information is published.

9. Portfolio Evidence

Screenshots should show actual inputs, outputs, matching evidence, hallucination behavior, paired fairness results, evaluation records and the human-review control. Crop unrelated content and never publish real candidate details.